



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic decisions

The Data Behind the Insights

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per transaction

37

Missing Values

In review ratings only

Customer Demographics

- Age, Gender, Location
- Subscription Status

Purchase Details

- Item, Category, Amount
- Season, Size, Color

Shopping Behavior

- Discounts & Promo Codes
- Review Ratings & Shipping

Python Analysis Pipeline

01

Data Loading & Exploration

Imported dataset, checked structure and summary statistics

02

Missing Data Handling

Imputed review ratings using median by category

03

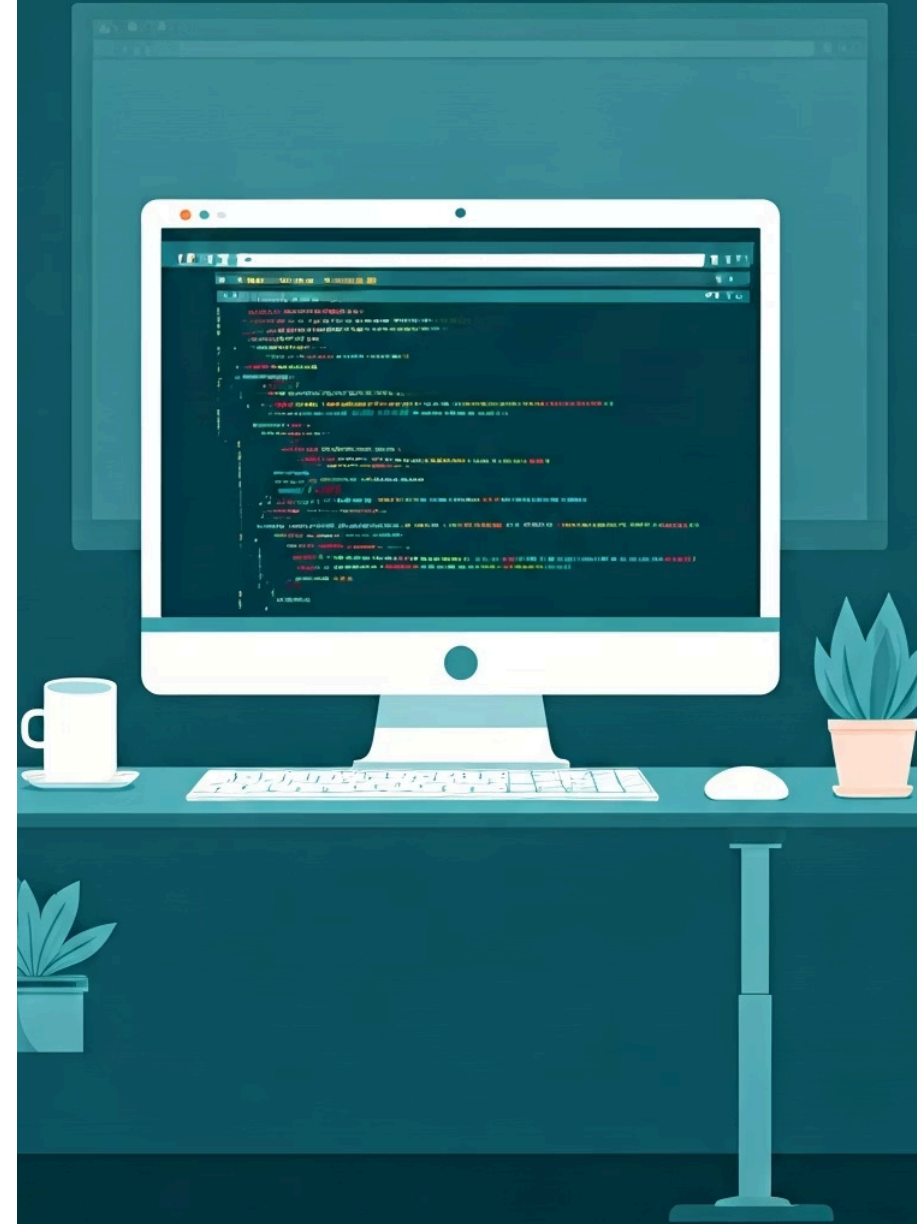
Feature Engineering

Created age groups and purchase frequency metrics

04

Database Integration

Connected to PostgreSQL for SQL analysis



Revenue by Gender



Key Finding

Male customers generated higher total revenue compared to female customers in the analyzed period



High-Value Discount Users

Identified customers who used discounts but still spent above average purchase amount

Strategic Insight

Discounts attract high-spenders who would purchase regardless

Opportunity

Optimize discount strategy to protect margins

Top-Rated Products Drive Satisfaction

Highest Average Ratings

1. Dress - Premium quality
2. Blouse - Consistent reviews
3. Sweater - Customer favorite
4. Shirt - Reliable choice
5. Pants - Strong performer



Shipping & Subscription Insights



Express Shipping

Higher average purchase amounts - premium customers prefer speed



Subscribers Spend More

Subscription customers show higher average spend and total revenue

CUSTOMER SEGMENTATION

Understanding Customer Loyalty



New Customers

First-time buyers
exploring products



Returning

Customers making
2-5 purchases



Loyal

5+ purchases -
most valuable
segment

Repeat buyers with 5+ purchases show significantly higher subscription rates



INTERACTIVE DASHBOARD

Power BI Visualization

Interactive dashboard enables real-time exploration of customer behavior patterns, revenue trends, and product performance metrics



Strategic Recommendations

1

Boost Subscriptions

Promote exclusive benefits to convert high-frequency buyers

2

Loyalty Programs

Reward repeat buyers to accelerate movement into loyal segment

3

Optimize Discounts

Balance sales growth with margin protection strategies

4

Product Positioning

Highlight top-rated items in marketing campaigns

5

Targeted Marketing

Focus on high-revenue age groups and express shipping users